SafeStep: A Low-Cost IoT-Enabled Semi-Autonomous Smart Walker for Blind and Elderly Navigation with Obstacle Avoidance, Water Hazard Detection, and Emergency Assistance

Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Science in Computer Science and Engineering of the University of Asia Pacific

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Declaration

We, hereby, declare that the work presented in this Thesis is the outcome of the investigation performed by us under the supervision of Nazmun Nahid, Assistant Professor, Department of Computer Science, University of Asia Pacific. We also declare that no part of this Thesis and thereof has been or is being submitted elsewhere for the award of any degree or Diploma.

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Also I am grateful to my family; who are, always with me in my every step of life.

Abstract

The demand for affordable assistive navigation technologies is rising every single day due to the growth in elderly population and the various problem / challenges faced by the visually impaired people. Also the traditional solutions such as walking sticks or walkers provide very few autonomous or semiautonomous mechanisms for obstacle avoidance and emergency communication. So, as a solution we suggest our project SafeStep which is a IoT-enabled semi-autonomous smart walker aimed to prioritize navigation independence for blind or elderly people. The SafeStep system uses an ESP32 microcontroller, three ultrasonic sensors, a water detection sensor, dual buzzer feedback mechanisms, DC motor-assisted navigation, and cloud-based emergency communication through the Blynk IoT platform. The main functionalities of the walker may be summarized as continuously monitoring its surroundings, detecting obstacles in multiple directions, identifying wet or slippery surfaces and providing directional guidance using sound alerts. Moreover, the emergency SOS feature allows the user to instantly send notification to his or her caregiver. A prototype was built and was tested in various environments to test its ability in obstacle detection, water hazard detection, navigation assistance, and emergency notifications. And the results showed that SafeStep prototype approximately correctly detected 90% of the obstacles correctly within the range of 5 cm to 150 cm and the optimal performance was observed between 20 cm to 80 cm. SafeStep was able to reduce the response time below one second and while delivering SOS notifications the response time was approximately one to two seconds. The proposed solution showed that embedded systems and IoT technologies can be effectively used to provide practical mobility assistance for elderly and visually impaired individuals at low-cost. SafeStep aims to work as a scalable foundation for the future of assistive robots in Bangladesh.

Keywords: Smart Walker, Assistive Robotics, IoT, ESP32, Obstacle Detection, Elderly Assistance, Blind Navigation, Ultrasonic Sensors, Emergency Notification, Embedded Systems

Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Background and Motivation

Mobility is a fundamental need for independence, confidence, and quality of life for both elderly and people with visual problems. According to global healthcare studies, the most common causes of injury for elderly and visually impaired population is falls and navigation related issues. Traditional sticks and walkers provide zero to none support in hazard detection or environment awareness.

Recent advancements in embedded systems, Internet of Things (IoT), robotics, and sensor technologies have enabled the development of intelligent assistive devices capable of providing real-time environmental awareness and user assistance. Smart mobility aids can enhance navigation by detecting obstacles, identifying hazards, and communicating emergency situations to caregivers or family members.

Despite significant progress in assistive technology, many commercially available smart walkers remain expensive and inaccessible in developing countries such as Bangladesh. Furthermore, existing solutions often focus on a single functionality such as obstacle detection or GPS tracking, while overlooking additional safety concerns including wet surface detection and emergency communication.

To address these limitations, this project proposes SafeStep, a low-cost IoT-enabled semi-autonomous smart walker designed to assist blind and elderly individuals through obstacle avoidance, water hazard detection, directional audio guidance, and emergency SOS notification capabilities.

1.2 Problem Statement

Various navigation related challenges both in indoor and outdoor environment is faced by the blind and elderly people in our country. Traditional method lack the ability to detect obstacles before any sort of physical contact which increases the possibility of collisions, falls and injuries.

Many existing smart assistance systems have tried to address this problem through obstacle detection mechanisms but most of these solutions have one or more of the following limitations: High implementation cost, sensing coverage limits, lackings in water hazard detection, absence of emergency communication features, complex architectures unsuitable for developing regions and lastly high power consumption.

The absence of a low-cost, multi-functional, and easily deployable assistive navigation system motivates the development of SafeStep.

1.3 Objectives of the Project

The main objective of our project is to design and implement a low-cost effective smart walker that increases the safety and navigational independence of elderly and visually impaired individuals.

The more specific objectives of our goals would be to design a semi-autonomous smart walker prototype, to detect obstacles using multiple ultrasonic sensors, to allow autonomous directional navigation, to detect wet or slippery surfaces using a water sensor, to provide audio directional feedback using buzzers, to add a IoT-based emergency notification mechanism, to develop a low-cost solution suitable for developing countries such as Bangladesh and lastly to evaluate system performance under real-world operating conditions.

1.4 Research Questions / Hypotheses

The project is guided by the following research questions:

RQ1: Can a low-cost smart walker provide reliable obstacle detection for blind and elderly users?

RQ2: Can ultrasonic sensor-based navigation assistance improve user safety during movement?

RQ3: How effectively can water hazard detection reduce environmental risks?

RQ4: Can IoT-based emergency notification improve caregiver awareness during emergencies?

RQ5: Is it possible to develop an affordable assistive navigation platform suitable for deployment in resource-constrained environments?

1.5 Scope of the Work

So we can clearly identify the scope of this project to :

- Design and construction of a semi-autonomous walker prototype.
- Integration of ESP32 microcontroller.
- Obstacle detection using three HC-SR04 ultrasonic sensors.
- Water hazard detection using a water sensor.
- Audio alert generation through dual buzzers.
- IoT integration through Blynk cloud services.
- Emergency SOS communication mechanism.
- Experimental testing and performance evaluation.

And lastly we want to highlight the fact that the project does not include: Computer vision systems nor GPS navigation nor AI-based object classification nor Medical monitoring systems and lastly we may include that this project is not a Commercial-grade mechanical construction.

1.6 Expected Outcomes

By the end of our project we aim to successfully develop a functional smart walker prototype which will have reliable obstacle detection capability along with real-time water hazard detection, effective directional guidance system, cloud-based emergency alert mechanism and improved user safety and mobility assistance which will lead to a successful demonstration of a scalable assistive robotics platform.

1.7 Impacts of the Project

1.7.1 Impact on Societal and Cultural Issues

The proposed system aims to improve accessibility and independence for vulnerable populations including elderly individuals and visually impaired users. Increased mobility can contribute to improved social participation, confidence, and quality of life.

1.7.2 Impact on Health, Safety, and Legal Issues

The system contributes to fall prevention through proactive obstacle and hazard detection. Emergency notification functionality enables timely communication with caregivers during potentially dangerous situations.

1.7.3 Impact on Environment and Sustainibility Issues

SafeStep utilizes low-power embedded hardware and inexpensive electronic components. The system demonstrates efficient utilization of computational resources while maintaining affordability and sustainability.

1.8 Report Organization

This report is organized into eight chapters.

Chapter 1 introduces the research problem, objectives, scope, and motivation.

Chapter 2 presents theoretical background and literature review.

Chapter 3 discusses system analysis, architecture, and design considerations.

Chapter 4 describes methodology, implementation, and technical development.

Chapter 5 presents experimental evaluation and performance analysis.

Chapter 6 discusses project planning and budget estimation.

Chapter 7 examines standards compliance, design constraints, and engineering accreditation requirements.

Chapter 8 concludes the study and outlines future research opportunities.

Chapter 2

Background and Literature Review

2.1 Preliminaries

Assistive robotics is an interdisciplinary field that combines robotics, embedded systems, sensing technologies, and intelligent control mechanisms to improve the quality of life of individuals with physical or sensory impairments.

2.1.1 Internet of Things (IoT)

The Internet of Things refers to interconnected physical devices capable of sensing, processing, and exchanging information through communication networks.

2.1.2 Ultrasonic Distance Measurement

Ultrasonic sensors measure distance by transmitting high-frequency sound waves and calculating the time required for reflected signals to return.

The distance calculation is given by:

$$Distance = \frac{(Speed of Sound \times Time)}{2} \tag{2.1}$$

2.1.3 Assistive Navigation Systems

Assistive navigation systems help users avoid hazards and navigate safely using environmental sensing technologies.

2.1.4 Embedded Systems

An embedded system combines hardware and software components designed for dedicated functionalities within larger systems.

2.1.5 Blynk IoT Platform

Blynk is a cloud-based IoT platform that enables remote monitoring, control, and event notification for embedded devices.

2.2 Related Works / Existing Systems

Most of the systems that we read have a core component in common which is the ultrasonic sensor. By using it they gained obstacle detection ability but the stick or walkers lack the autonomous ability of navigation support. Also the autonomous walkers are seen to use advanced technologies such as LiDAR, Camera or even machine learning algorithms but even though they provide high accuracy they also increase the implementation cost by a lot. Which in SafeStep we tried to avoid doing. Some IoT devices are seen to have hazard detection such as wet floor recognition but a majority of them lack that. From the review we come to know that the most of the existing systems have few uniques features of their own but they all could be put together to produce something highly efficient which we aim to do.

2.3 Comparative Analysis of Related Works

Table 2.1: Comparative Analysis of Existing Assistive Navigation Systems

Study	**Obstacle Detection**	**Water Detection**	**Emergen Alert**	**IoT Support**	**Cost**
[6] Mobility Support with Intelligent Obstacle Detection for Enhanced Safety	Yes	No	No	Yes	Medium
[5] Smart Stick Navigation System for Visually Impaired Based on Machine Learning Algorithms Using Sensors Data	Yes	Yes	No	Yes	High
[7] A Smart Cane Based on 2D LiDAR and RGB-D Camera Sensor Realizing Navigation and Obstacle Recognition	Yes	No	No	Yes	High
Proposed SafeStep	Yes	Yes	Yes	Yes	Low

2.4 Research Gap / Limitations of Existing Methods

From the literature review we clearly identified many traditional similarities between the existing systems but the main limiting factors that we aim to improve was also discovered such a having a high implementation cost, no or very little hazard detection, the lacking of safety mechanisms or having a limited emergency communication support.

But we aim to achieve all of the above using our project SafeStep by implementing a low cost design, multiple sensor based navigation system, water hazard detection, SOS alert and a overall light weight and affordable design.

2.5 Summary

This chapter reviewed the theoretical foundations of assistive robotics, embedded systems, ultrasonic sensing, and IoT technologies. Existing studies demonstrate the effectiveness of intelligent mobility aids but reveal several limitations regarding affordability, hazard detection, and integrated safety functionality. The identified research gaps justify the development of SafeStep as a low-cost, multi-functional smart walker capable of providing obstacle avoidance, water hazard detection, and emergency communication assistance within a single platform.

Chapter 3

System Analysis and Design

3.1 Requirement Analysis

The system requirements were identified through analysis of mobility challenges faced by elderly individuals and visually impaired users, observations of conventional mobility aids, and review of related assistive navigation systems.

3.1.1 Stakeholder Analysis

The primary stakeholders include:

Primary Users

- Visually impaired individuals
- Elderly individuals
- Users with limited mobility

Secondary Users

- Family members
- Caregivers
- Healthcare assistants

Technical Stakeholders

- System developers
- Maintenance personnel
- Future researchers

3.1.2 Functional Requirements

Table 3.1: Functional Requirements

ID	Requirement Description
FR1	Detect obstacles in front of the walker
FR2	Detect obstacles on the left side
FR3	Detect obstacles on the right side
FR4	Automatically select safer navigation direction
FR5	Detect wet or slippery surfaces
FR6	Stop movement upon water hazard detection
FR7	Generate directional buzzer alerts
FR8	Send SOS notifications through Blynk
FR9	Enable ON/OFF control through push button
FR10	Provide semi-autonomous navigation support

3.1.3 Non-Functional Requirements

Table 3.2: Non-Functional Requirements

ID	Requirement Description
NFR1	Low-cost implementation
NFR2	Lightweight construction
NFR3	Response time below 1 second
NFR4	Reliable wireless communication
NFR5	User-friendly operation
NFR6	Low power consumption
NFR7	Easy maintenance
NFR8	Expandable architecture

3.2 System Overview / Architecture

SafeStep follows a layered architecture consisting of sensing, processing, communication, and actuation layers.

3.2.1 High-Level Architecture

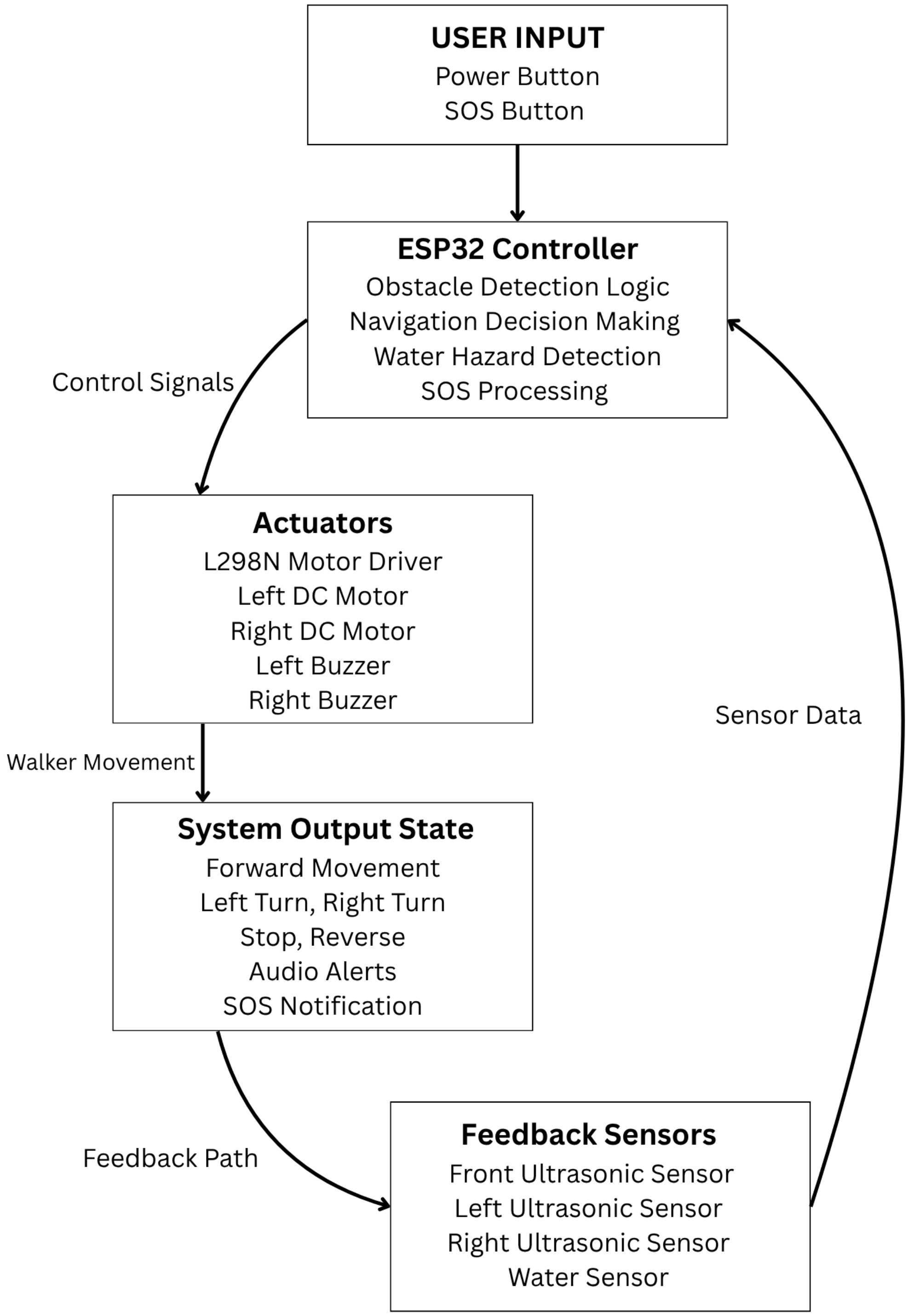

Figure 3.1: SafeStep Control System Diagram

3.2.2 Detailed Design Diagrams

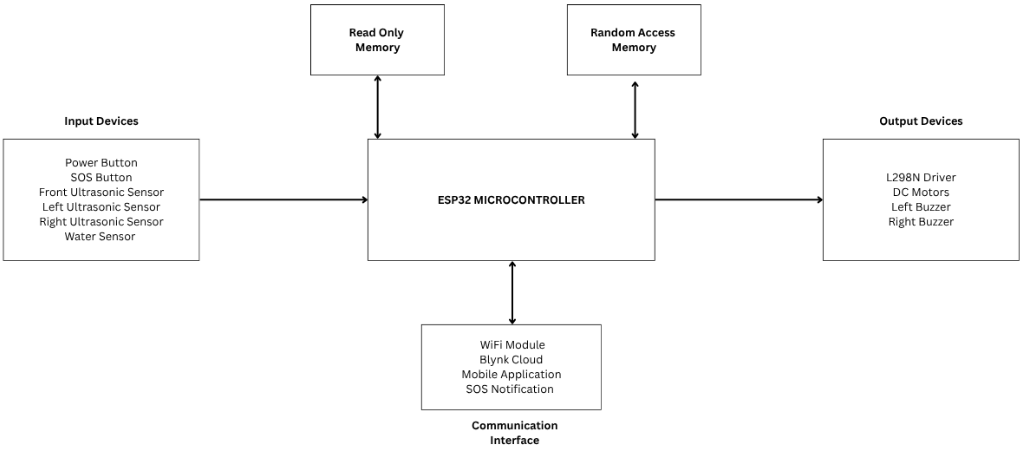

Figure 3.2: Hardware Achitecture of SafeStep Embedded System

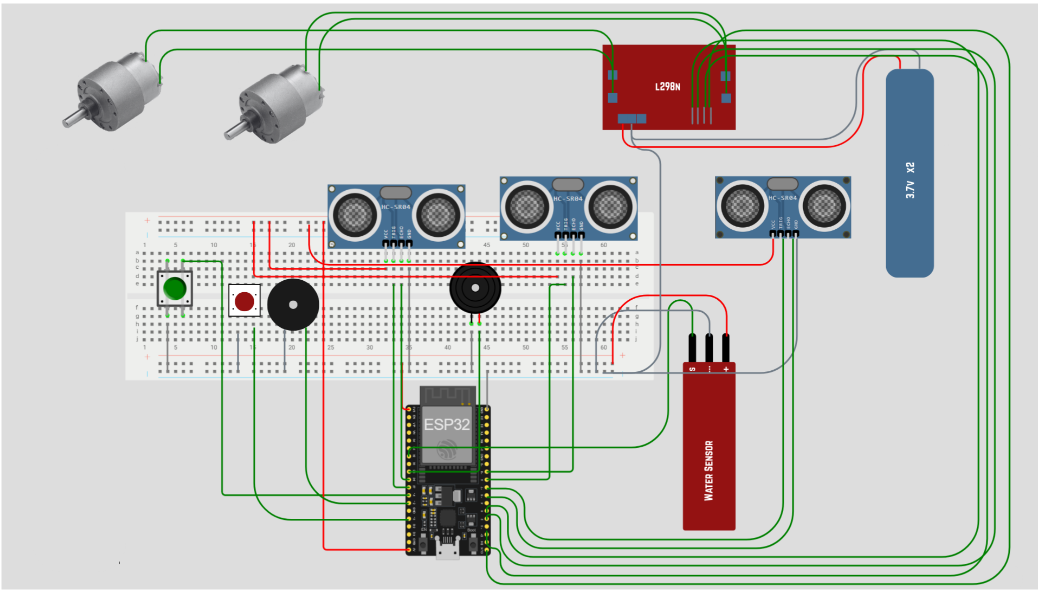

Figure 3.3: Circuit Diagram

3.2.3 Design Alternatives and Rationale

Several design alternatives were evaluated before selecting the final architecture.

Table 3.3: Obstacle Detection Technologies

Technology	Cost	Complexity
Camera	High	High
LiDAR	Very High	High
Ultrasonic Sensor	Low	Low

Ultrasonic sensing was selected due to affordability and simplicity.

Table 3.4: Controller Selection

Controller	Cost	WiFi
Arduino Uno	Medium	No
Raspberry Pi	High	Yes
ESP32	Low	Yes

ESP32 was selected due to having a integrated WiFi, low cost, sufficient GPIO pins, low power consumption.

3.3 Algorithmic Flow and Pseudocode

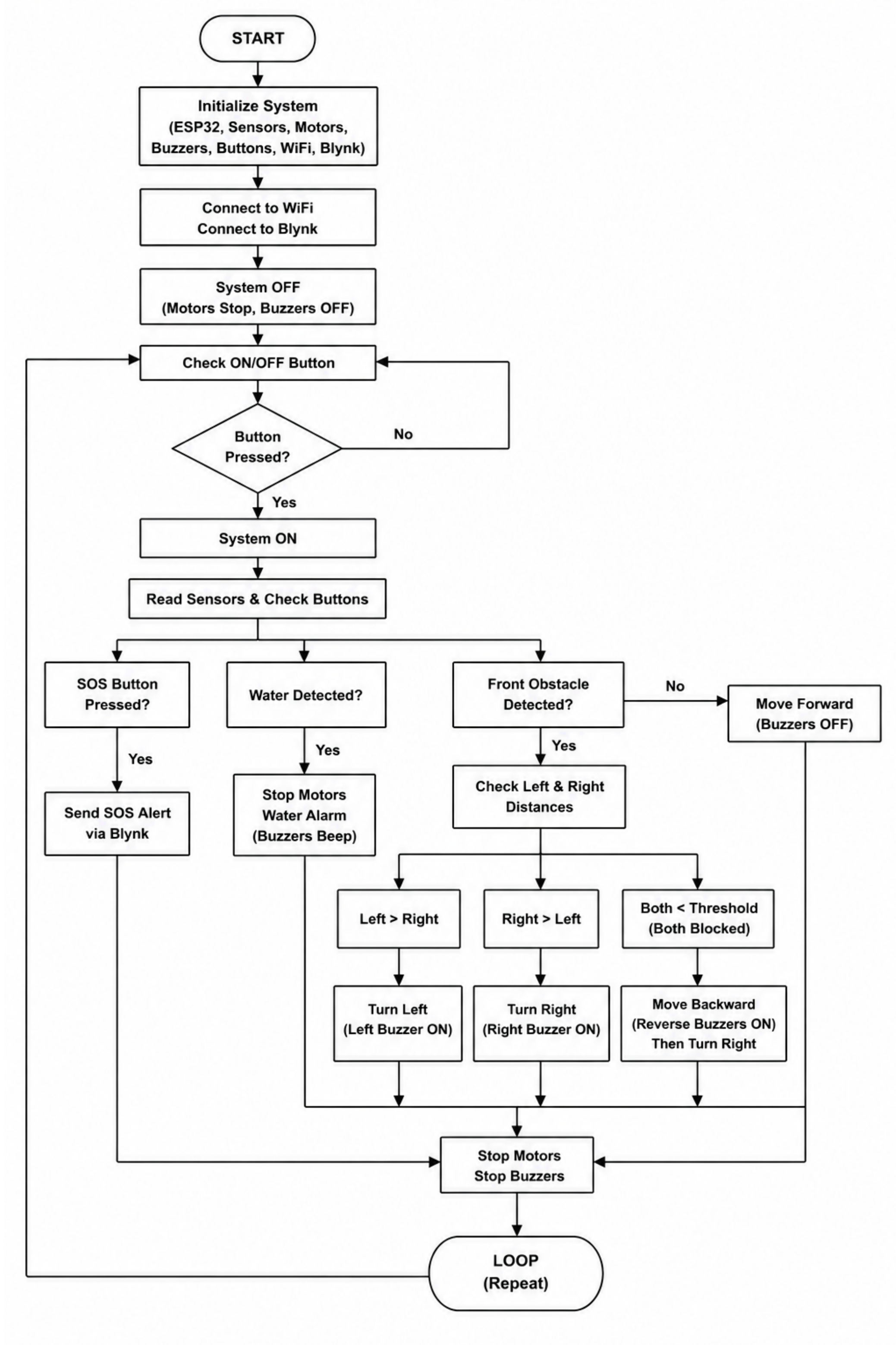

Figure 3.4: SafeStep Flowchart

The navigation algorithm continuously monitors environmental conditions and determines appropriate actions.

Pseudocode

```
INITIALIZE esp32

INITIALIZE sensors

INITIALIZE blynk

WHILE system running

  CHECK SOS button

  IF SOS pressed

    SEND blynk notification

  ENDIF

  CHECK main power button

  IF OFF

    STOP motors

    STOP buzzers

    continue

  ENDIF

  READ water sensor

  IF water DETECTED

    STOP motors

    ACTIVATE water alarm
```

```
    continue
ENDIF
READ front distance
IF front clear
    MOVE forward
ELSE
    STOP motors
    READ left distance
    READ right distance
    IF both blocked
        MOVE backward
        TURN right
    ELSE IF left clear
        TURN left
    ELSE
        TURN right
    ENDIF
ENDIF
```

Navigation Decision Function

The walker selects navigation direction based on:

$$Direction = \begin{cases} Left & \text{if} & left\,is\,clear \\ Right & \text{if} & right\,is\,clear \\ Reverse & \text{otherwise} & \end{cases} \tag{3.1}$$

3.4 Summary

This chapter presented the requirements analysis, system architecture, hardware design, and navigation algorithm of the SafeStep smart walker. The modular architecture enables efficient interaction among sensing, processing, communication, and actuation modules. The design emphasizes affordability, simplicity, and user safety while maintaining expandability for future enhancements.

Chapter 4

Methodology and Implementation

4.1 Methodological Framework

The project followed a structured engineering development methodology consisting of seven phases.

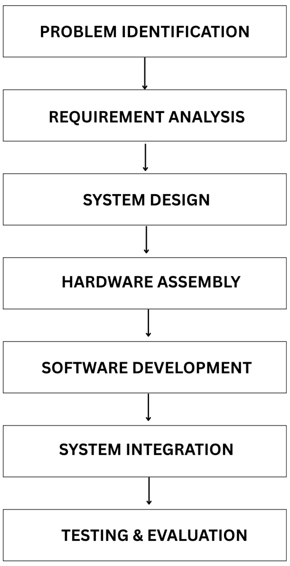

Figure 4.1: Project Development Framework

4.2 Tools, Libraries, and Technologies Used

Table 4.1: Development Technologies

Category	Technology
Programming Language	C++
IDE	Arduino IDE
Microcontroller	ESP32
Cloud Platform	Blynk
Communication	WiFi
Sensor Technology	Ultrasonic + Water Sensor
Motor Control	L298N

Software Libraries

Table 4.2: Libraries Used

Library	Purpose
WiFi.h	WiFi communication
WiFiClient.h	Client networking
BlynkSimpleEsp32.h	IoT notification support

4.3 Data Collection / Dataset Description

Since this is an IoT and robotics project, data was collected through sensors rather than datasets.

Sensor Data Types

- Front Ultrasonic sensor gave us Front distance

- Left Ultrasonic sensor gave us Left distance
- Right Ultrasonic sensor gave us Right distance
- Water sensor gave us Moisture level
- Buttons gave us User input

4.4 Preprocessing Steps

The following preprocessing operations were implemented to get a better and acceptable output : noise filtering through repeated sensing, distance threshold comparison, water sensor threshold calibration, debounce handling for buttons, invalid sensor reading rejection.

Threshold Values

Table 4.3: Threshold Values

Parameter	**Value**
Obstacle Distance	20 cm
Water Threshold	200
Button Debounce	50 ms

4.5 Model Development / Algorithm Design

Unlike machine learning systems, SafeStep utilizes a rule-based decision model.

Decision Logic

1. When Front is Clear, Move Forward
2. When Obstacle is Detected, Check Side Sensors

3. When Left is More Open, Turn Left
4. When Right is More Open, Turn Right
5. When Both are Blocked then Reverse + Turn
6. When Water is Detected, Stop
7. When SOS is Pressed, Send Alert

4.6 System Implementation / Modules Description

The implementation consists of five major modules.

Module 1: Obstacle Detection Module Uses three HC-SR04 sensors to measure surrounding distances. Functions: Front scanning, Left scanning, Right scanning.

Module 2: Navigation Module Determines movement direction. Capabilities: Forward movement, Left turn, Right turn, Reverse recovery.

Module 3: Water Hazard Module Monitors moisture levels. Functions: Surface analysis, Hazard detection, Movement stopping

Module 4: Audio Feedback Module Provides directional information. Conditions and their buzzers : When Left Turn, Left Buzzer is activated and When Right Turn, Right Buzzer is activated and While reversing, Both Buzzers are activated and When Water is detected Both Buzzers give Pulses.

Module 5: IoT Emergency Module Features: WiFi connectivity, Blynk integration, SOS notification

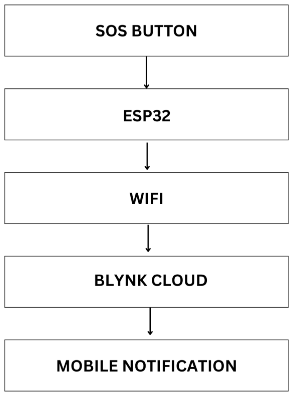

Figure 4.2: Blynk Communication Workflow

4.7 Hardware and Software Requirements

Hardware Requirements

Table 4.4: Hardware Components

Component	Quantity
ESP32	1
HC-SR04	3
Water Sensor	1
L298N	1
DC Motor	2
Buzzer	2
Push Button	2
Battery	2
Walker Chassis	1

Software Requirements

Table 4.5: Software Requirements

Software	Version
Arduino IDE	2.x
ESP32 Board Package	Latest
Blynk IoT Platform	Latest
Windows OS	Windows 10/11

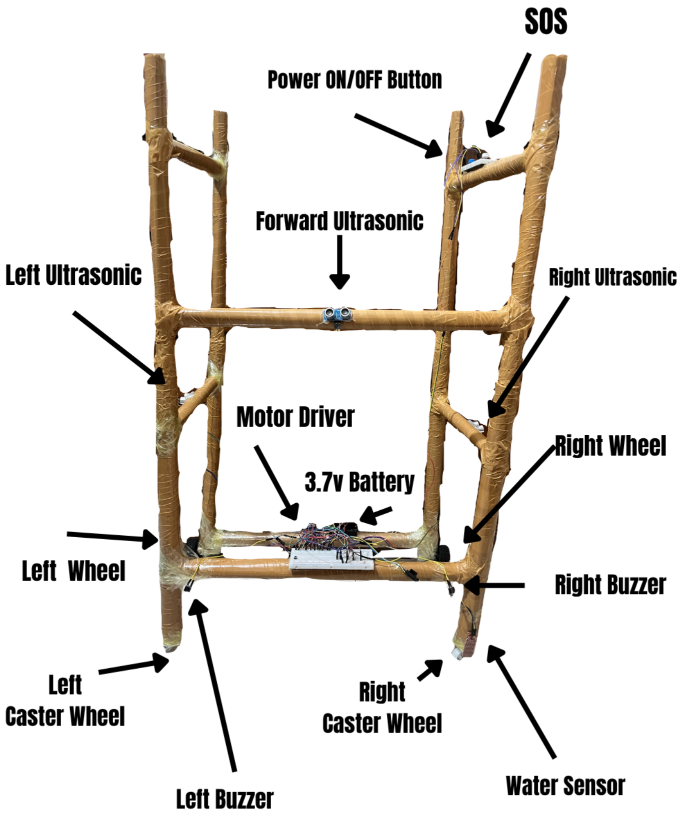

Figure 4.3: Prototype Photograph

4.8 Summary

This chapter described the development methodology, implementation process, hardware architecture, software framework, and system modules of the SafeStep smart walker. The integration of sensing, navigation, IoT communication, and safety mechanisms establishes a complete assistive mobility platform suitable for elderly and visually impaired users.

Chapter 5

Results and Evaluation

5.1 Experimental Setup

The prototype was physically assembled and tested under indoor and semi-controlled environmental conditions.

The evaluation focused on: Obstacle detection capability, Directional navigation assistance, Water hazard detection, SOS emergency notification, System response time, Battery performance, Hardware Setup.

The experimental setup consisted of: ESP32 microcontroller, Three HC-SR04 ultrasonic sensors, Water detection sensor, L298N motor driver, Two DC motors, Two buzzers, Two push buttons, Blynk mobile application, Dual 3.7V battery supply.

5.1.1 Testing Scenarios

Table 5.1: Test Scenarios

Test Type	**Number of Tests**
Obstacle Detection	50
Navigation / Turning	30
Water Detection	20
SOS Notification	20
Total Tests	**120**

5.2 Performance Metrics

The following performance metrics were used. **Obstacle Detection Accuracy**

$$Accuracy = \frac{CorrectDetections}{TotalTests} \times 100 \tag{5.1}$$

Water Detection Reliability

$$Reliability = \frac{SuccessfulWaterDetections}{TotalWaterTests} \times 100 \tag{5.2}$$

Notification Delay

Average time required for notification delivery.

System Response Time

Time between obstacle detection and navigation decision.

5.3 Result Analysis

5.3.1 Obstacle Detection Performance

The obstacle detection module achieved strong performance across different obstacle positions and distances.

Table 5.2: Obstacle Detection Results

Parameter	Value
Total Tests	50
Successful Detections	45
Detection Accuracy	90%
Detection Range	5–150 cm
Optimal Range	20–80 cm
Response Time	< 1 second

The results indicate that ultrasonic sensing provides sufficient environmental awareness for assistive navigation applications.

Figure 5.1: Obstacle Detection Accuracy Chart

5.3.2 Water Hazard Detection Performance

The water detection module was evaluated using multiple wet surface scenarios.

Table 5.3: Water Detection Results

Parameter	Value
Tests Conducted	20
Successful Detections	14
Reliability	70%

The lower accuracy compared to obstacle detection can be attributed to sensor sensitivity variations and environmental moisture conditions.

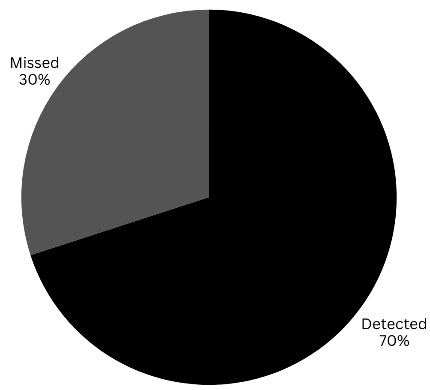

Figure 5.2: Water Detection Reliability

5.3.3 SOS Notification Performance

The emergency communication feature demonstrated reliable operation.

Table 5.4: SOS Notification Results

Parameter	Value
Total Tests	20
Successful Notifications	20
Success Rate	100%
Average Delay	1–2 Seconds

The Blynk cloud platform successfully delivered notifications with minimal latency.

5.3.4 Navigation Performance

The walker was tested in various obstacle arrangements.

Table 5.5: Navigation Evaluation

Scenario	Performance
Straight Path	Successful
Left Obstacle	Turn Right
Right Obstacle	Turn Left
Both Sides Open	Chosen Safest Route
Front Blocked	Avoidance Successful
Fully Blocked	Reverse Recovery Successful

The navigation algorithm successfully performed semi-autonomous movement decisions.

5.3.5 Battery Performance

Table 5.6: Battery Analysis

Parameter	**Value**
Battery Type	2×3.7V
Configuration	Direct Supply
Runtime	Approximately 20 Minutes
Motor Condition	Underpowered Operation

The limited battery runtime represents one of the primary constraints of the current prototype.

5.4 Comparison with Existing Approaches

A comparison was conducted against representative assistive navigation systems reported in recent literature.

Table 5.7: Comparative Analysis

Feature	**Smart Cane**	**Smart Stick**	**Advanced Walker**	**SafeStep**
Obstacle Detection	Yes	Yes	Yes	Yes
Water Detection	No	Limited	Limited	Yes
SOS Alert	No	Yes	Yes	Yes
IoT Support	No	Yes	Yes	Yes
Semi-Autonomous Navigation	No	No	Yes	Yes
Low Cost	Moderate	Moderate	No	Yes

The comparison demonstrates that SafeStep integrates multiple safety functionalities while maintaining affordability.

5.5 Discussion on Findings

The experimental results demonstrate that the proposed SafeStep system effectively addresses several mobility challenges faced by elderly and visually impaired users. Key findings include: High obstacle detection accuracy (90%), Fast reaction time (¡1 second), Reliable emergency notification capability, Effective directional guidance, Low implementation cost, Lightweight architecture.

The integration of multiple assistive functionalities into a single platform distinguishes SafeStep from many existing low-cost navigation aids.

The results also validate the feasibility of utilizing low-cost embedded systems and IoT technologies for assistive healthcare applications in developing countries.

5.6 Limitations

Despite encouraging results, several limitations were observed. Few Hardware Limitations were observed such as Limited battery capacity, underpowered motor operation, dependence on WiFi availability. Also few Sensor Limitations were also visible such as ultrasonic blind spots, reduced water detection reliability, environmental noise sensitivity. And the future works may include GPS tracking, computer vision, object classification, No voice interaction. These limitations provide opportunities for future improvements.

5.7 Summary

This chapter evaluated the performance of the SafeStep prototype through multiple experimental scenarios. Results demonstrated 90% obstacle detection accuracy, 70%

water hazard detection reliability, and successful emergency communication capabilities. The findings confirm the viability of SafeStep as a low-cost assistive mobility platform.

Chapter 6

Project Planning and Budget

6.1 Project Timeline

A structured development approach was followed throughout the project lifecycle.

The project was completed over seven weeks.

Table 6.1: Seven-Week Project Timeline

Activities	**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8**
Literature Review								
Requirement Analysis								
System Design								
Hardware Procurement								
Circuit Development								
Software Development								
Prototype Integration								
Testing & Validation								
Documentation								
Final Presentation								

6.2 Budget Estimation

The total project budget was kept intentionally low to maintain affordability and accessibility.

Table 6.2: Hardware Cost Breakdown

Component	**Quantity**	**Unit Cost (BDT)**	**Total Cost (BDT)**
ESP32	1	600	600
HC-SR04 Sensor	3	100	300
L298N Driver	1	300	300
DC Motors	2	500	1000
3.7V Batteries	2	200	400
Buzzers	2	100	200
Walker Chassis Materials	1	500	500
Miscellaneous Components	–	1000	1000
		Total Estimated Cost	**4300 BDT**

Table 6.3: Cost Distribution

Category	**Percentage**
Motors	23.3%
Electronics	32.6%
Power System	9.3%
Chassis	11.6%
Miscellaneous	23.2%

6.3 Summary

This chapter presented the project timeline and budget estimation associated with the development of SafeStep. The seven-week development schedule ensured systematic project execution, while the low-cost budget of approximately 4300 BDT demonstrated the feasibility of creating affordable assistive navigation systems for developing regions.

Chapter 7

Standards and Design Constraints

7.1 Compliance with Standards and Professional Practice

The development of SafeStep followed established engineering principles related to embedded systems, IoT communication, assistive technology, and user safety.

The project emphasizes: Reliable sensing and control, User safety, Ethical use of technology, Low-power system design, Accessibility for vulnerable populations, Maintainable system architecture.

7.2 Design Constraints

Several constraints influenced the final design of the SafeStep system. **Power Constraints**

The prototype operates using Two 3.7V batteries.

This resulted in Limited runtime, Reduced motor torque, Lower navigation speed.

Sensor Constraints

Ultrasonic sensors introduce Blind zones, Surface reflection issues, Environmental sensitivity.

Communication Constraints

The SOS module requires Internet access, WiFi connectivity, Blynk cloud availability.

Budget Constraints

The project was designed under strict affordability requirements.

Total budget: 4300 BDT

This significantly influenced component selection decisions.

7.3 Complex Engineering Problem (CEP) Coverage

The SafeStep project addresses several attributes associated with Complex Engineering Problems. **CEP Mapping**

Table 7.1: CEP Mapping

CEP Attribute	How Addressed in Project	Evidence
P1: Depth of Knowledge	Embedded systems, IoT, robotics integration	Chapters 3 & 4
P2: Conflicting Requirements	Cost vs functionality trade-offs	Section 3.2.3
P3: In-depth Analysis	Sensor selection and navigation logic	Chapter 3
P4: Familiar and Unfamiliar Problems	Mobility assistance challenges	Chapters 1 & 2
P5: Wide-Ranging Issues	Safety, accessibility, affordability	Chapter 7
P6: Stakeholder Needs	Elderly and visually impaired users	Section 3.1
P7: Significant Consequences	User safety and mobility assistance	Chapter 5

CEP Justification

The project required multidisciplinary knowledge involving:

- Robotics
- Embedded systems
- Electronics
- IoT communication
- Human-centered design

The integration of these domains demonstrates the complexity expected in engineering problem solving.

7.4 Complex Engineering Activities (CEA) Coverage

The project also satisfies several Complex Engineering Activity requirements. **CEA Mapping**

Table 7.2: CEA Mapping

CEA Attribute	**How Addressed**	**Evidence**
A1: Diverse Resources	Hardware, software, cloud services	Chapters 3 & 4
A2: Resolution of Interactions	Sensor-actuator coordination	Chapter 4
A3: Creative Innovation	Combined obstacle, water, and SOS system	Chapter 2
A4: Consequences to Society	Mobility assistance and safety	Chapter 1
A5: Engineering Judgment	Component selection and optimization	Chapter 3

7.5 Summary

This chapter demonstrated compliance with engineering standards, ethical responsibilities, sustainability considerations, and accreditation requirements. The CEP and CEA mappings confirm that SafeStep addresses a complex engineering problem through the application of multidisciplinary engineering knowledge and innovative system development.

Chapter 8

Conclusion and Future Work

8.1 Summary of the Work

Restates objectives and summarizes the project achievements. This project presented the design, implementation, and evaluation of SafeStep, a low-cost IoT-enabled semi-autonomous smart walker intended to assist elderly and visually impaired users during navigation. The system integrates ESP32 microcontroller, Three ultrasonic sensors, Water hazard sensor, Dual directional buzzers, Motorized navigation support, Blynk-based SOS communication.

The developed prototype demonstrates the feasibility of combining embedded systems, sensing technologies, and IoT communication into a practical assistive mobility solution.

8.2 Key Findings and Contributions

The primary contributions of this research include:

C1: Multi-Hazard Detection

The system detects Obstacles, Water hazards, Emergency conditions within a unified architecture.

C2: Affordable Design

The complete prototype was developed at approximately: 4300 BDT making it significantly cheaper than many commercial alternatives.

C3: IoT Integration

The SOS functionality successfully enables Real-time emergency notification, Caregiver awareness, Remote assistance.

C4: Semi-Autonomous Navigation

The walker demonstrates Automatic obstacle avoidance, Directional decision making, Safe route selection.

C5: Real-World Validation The prototype was physically built and experimentally tested. Key results include:

- 90% obstacle detection accuracy
- 70% water detection reliability
- $<$ 1 second response time
- 1–2 second SOS notification delay

8.3 Limitations of the Study

Despite promising results, several limitations remain. Hardware Limitations are Low battery runtime, Underpowered motors, Lightweight prototype frame and Sensor Limitations are Ultrasonic blind zones, Environmental interference. Lastly Functional Limitations are No GPS functionality, No voice assistance, No object recognition, No machine learning capability.

8.4 Recommendations and Future Work

Several improvements can enhance future versions of SafeStep.

GPS Tracking

Integration of GPS modules can enable Real-time location tracking, Route monitoring, Emergency location sharing.

AI-Based Object Recognition Future systems may incorporate Computer vision, Object classification, Scene understanding using lightweight AI models.

Voice Assistance Voice interaction can improve usability through Audio navigation instructions, Voice commands, Emergency voice alerts.

Advanced Power Management

Future enhancements may include Rechargeable battery packs, Solar-assisted charging, Battery health monitoring.

Mobile Application Development

A dedicated mobile application can provide Live walker status, Navigation history, Remote caregiver monitoring.

Healthcare Integration

Potential future features include Heart rate monitoring, Fall detection, Health analytics.

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Appendices

A Source Code / GitHub Link

GitHub Repository: https://github.com/MD-Sydul-Islam-Zubair/Smart-Walker

B Team Responsibilities

MD. Sydul Islam Zubair

- Complete software development
- ESP32 programming
- Blynk integration
- Motor driver interfacing
- Hardware integration
- Project supervision
- Project Report

Suna Hera Akter Suna

- Documentation support
- Testing assistance
- Literature review support

- Chassis creation
- Assisted in Project Report

Farhana Haque Mahima

- Conducted Survey 1
- Conducted Survey 2

C Survey 1

SQ1: Your Name

Ans: Delowara begum.

SQ2: What is your job role?

Ans: Retired.

SQ3: Are you involved in purchasing/decision making?

Ans: No.

SQ4: Have you understood the product concept?

Ans: 5 out of 5.

SQ5: Do you think our product will be helpful for your workplace?

Ans: 5 out of 5.

Additional comments:

This is a very good initiative for our elderly people.

D Survey 2

SQ1: Your Name

Ans: Jamila Khatun.

SQ2: What is your job role?

Ans: Retired.

SQ3: Are you involved in purchasing/decision making?

Ans: No.

SQ4: Have you understood the product concept?

Ans: 5 out of 5.

SQ5: Do you think our product will be helpful for your workplace?

Ans: 4 out of 5.

SQ6: Would you purchase this product?

Ans: 3 out of 5.

SQ7 Do you think the product price is acceptable?

Ans: 4 out of 5.

SQ8: How much money would you spend for this ind of product?

Ans: From 3000BDT- 5000BDT

SQ9: Which part of this product needs improvement?

Ans: The battery life should be improved because it runs out quite quickly from what I heard. The buzzer volume could also be adjustable for users with hearing difficulties.

Additional comments:

The walker will help me notice objects that I might not have seen otherwise. The emergency notification feature is also reassuring because family members can be alerted when needed. Overall, it felt useful and practical.

Figure 1: Collected Images while conducting survey 2